

CONFERENCE PREVIEW & PROJECT APPROVAL

Explainable and Calibrated CLV Prediction for VIP Targeting with Semantic Customer Profiling

Group 11 — FPT University

Course: ADY201m — AI &
Data Science

Language: Python &
SQL (DuckDB)

Presentation Agenda

Theoretical Motivation

 **Problem Motivation:** Why ranking is the ultimate VIP targeting objective.

 **Research Gaps:** Eliminating lookahead bias, understanding semantic taste.

Methodology & Results

 **Proposed Pipeline:** Walk-forward split, Hurdle methodology, Conformal intervals.

 **Key Results:** 17-model benchmark performance, ROI, and calibration metrics.

Why CLV Prediction Matters for *VIP Targeting*

- ✓ Retailers operate with highly constrained marketing budgets, necessitating efficient resource allocation.
- ✓ Instead of generic blasting, companies must identify and prioritize high-value customers for exclusive VIP incentives.
- ✓ VIP targeting shifts CLV prediction from a pure regression problem to a high-stakes **ranking optimization problem**.

CORE BUSINESS QUESTION:

“Which top 10% of customers will capture the maximum future revenue share?”

VIP TARGETING FUNNEL

 All Active Customers Database



 Robust CLV Ranking Model



 Curated Top 10% VIP Target Group



 Strategic Marketing Campaign Allocation

Satisfying *ADY201m* Academic Pipeline Demands

Formal Requirement	Project Implementation Details	Status & Academic Rigor
Python Science Pipeline	End-to-end robust ETL, cleaning, validation, modeling, and output.	✔ Fully Completed
SQL Analysis	In-memory processing with DuckDB for cohorts, RFM, and segmentations.	✔ Native Integration
≥ 5 Machine Learning Models	Evaluated 17 separate configurations across probabilistic and ML categories.	✔ Exceeded Core Goal
Research-Based Learning	Implementation of BG/NBD, Gamma-Gamma, ZILN, CQR, and SHAP.	✔ Springer Template Standard
Dashboard & Reports	Interactive Streamlit dashboard + 12-page comprehensive scientific paper.	✔ Ready for Production
AI Audit Log	Documented log featuring prompt structures, iterations, and resolution steps.	✔ Standard Adherence

Key Research Gaps Handled by *Our Framework*

1

Leakage-Free Validation

Temporal Evaluation Leakage

Standard static k-fold CV fails on transactional streams. Our framework introduces rigid, strictly partition-based walk-forward temporal windows to emulate zero-leakage production evaluation.

2

Semantic Behavior

Limits of Classic RFM

Traditional RFM captures transaction frequencies and monetary weights, but omits *what* customers purchase. We integrate semantic co-purchase graph structures to map underlying product tastes.

3

Calibrated Uncertainty

Point Prediction Risks

Using raw regression estimates risks capital efficiency in marketing budget targeting. We calibrate interval boundaries using Conformalized Quantile Regression (CQR) to assure 95% reliable coverage.

Database Overview: *Online Retail II*

- ✓ Dataset is sourced directly from the reputable **UCI Machine Learning Repository**.
- ✓ Captures all real-world transaction details from a prominent UK-based non-store online retailer.
- ✓ Spans a robust historical period tracking purchases made between **December 2009 and December 2011**.
- ✓ Highly suited for validation owing to detailed sequences of transaction IDs, stock IDs, timestamps, and customer accounts.

805,549

TRANSACTIONS

5,038

CUSTOMERS

41

COUNTRIES

\$17.7M

TOTAL REVENUE

🔍 **Preprocessing standard:** All accounts lacking a registered identifier were discarded to maintain sequence tracking integrity. **Median Spend: \$899**

The Challenge: *Zero-Inflated* & Heavy-Tailed CLV

Modeling transaction history is statistically challenging because of two primary mathematical characteristics:

- ✓ **Extreme Zero-Inflation:** The majority of buyers do not execute transactions within subsequent future horizons, producing spikes at \$0 spend labels.
- ✓ **Heavy Right-Tail Skewness:** A very small cohort of high-value purchasers generates massive relative returns, heavily pulling the distribution's mean.
- ✓ **Methodological Takeaway:** Traditional single-stage MSE regressions fail to learn effectively from these combined structures.

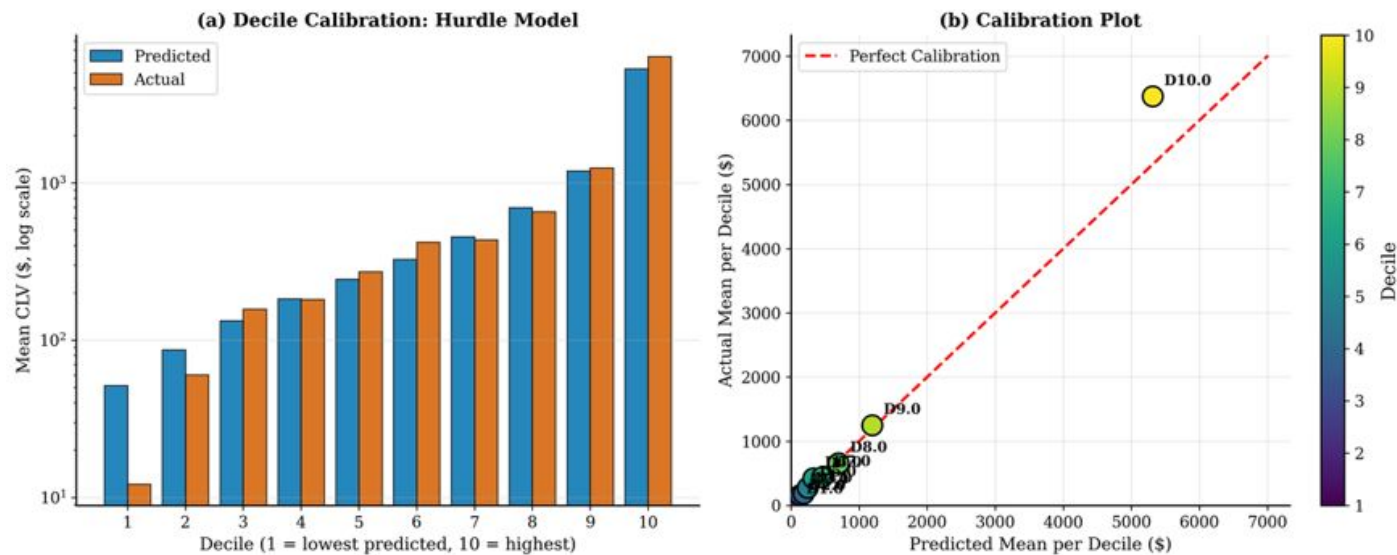
42% – 67%

FUTURE ZERO-SPEND RATES

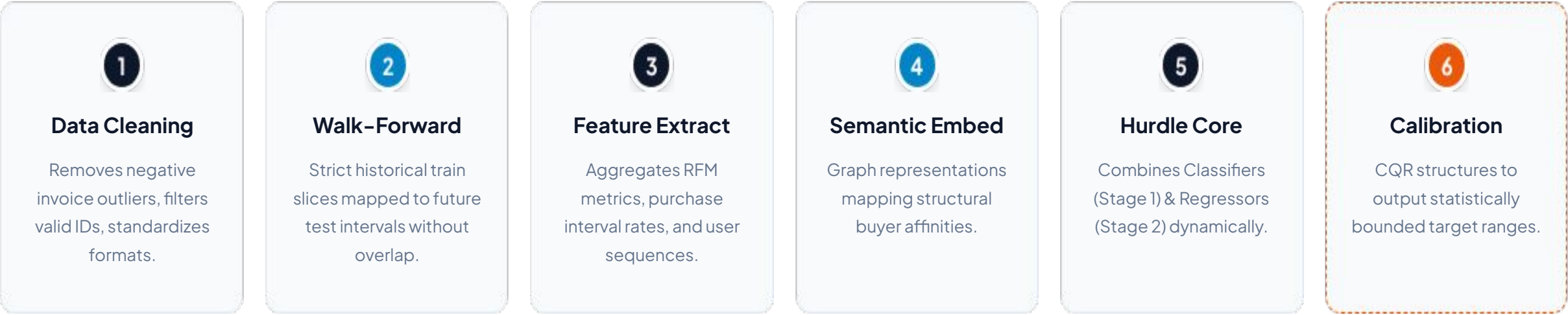
High frequency of dormant or single-time shoppers during prediction windows.

22.95

Figure 6: Calibration Analysis (Window 3, n=986 test customers)



Proposed *Deployment-Oriented* CLV Pipeline



Note: Pipeline ensures rigorous deployment separation where Stage 1 calculates $P(\text{Active})$ and Stage 2 computes $E(\text{Revenue} \mid \text{Active})$.

Validation Protocol: *Leakage-Free* Horizons

WINDOW	OBSERVATION HISTORY	HORIZON
W1	12 Months 3M Horizon	Main Bench
W2	15 Months 3M Horizon	Main Bench
W3 (Primary)	18 Months 6M Horizon	Deploy Target
W4	9 Months 3M Horizon	Sensitivity
W5	21 Months 3M Horizon	Sensitivity

- ✓ **Rigid Cutoffs:** Features strictly compute before the split; targets reside strictly after.
- ✓ **Primary Deployment:** Window W3 represents the main deployment simulation setting.
- ✓ **Robust Audits:** Windows W4 & W5 operate as sensitivity verification frames.

Systematic Benchmark of *17 Evaluated Models*



Baselines & Probabilistic

Simple Baselines: Historical Mean, Historical Monetary value, RFM Grid score.

Classical Frameworks: BG/NBD probability + Gamma-Gamma spending estimators.



Advanced Machine Learning

Linear & Trees: ElasticNet, Ridge, Extreme Gradient Boosting (XGBoost), LightGBM.

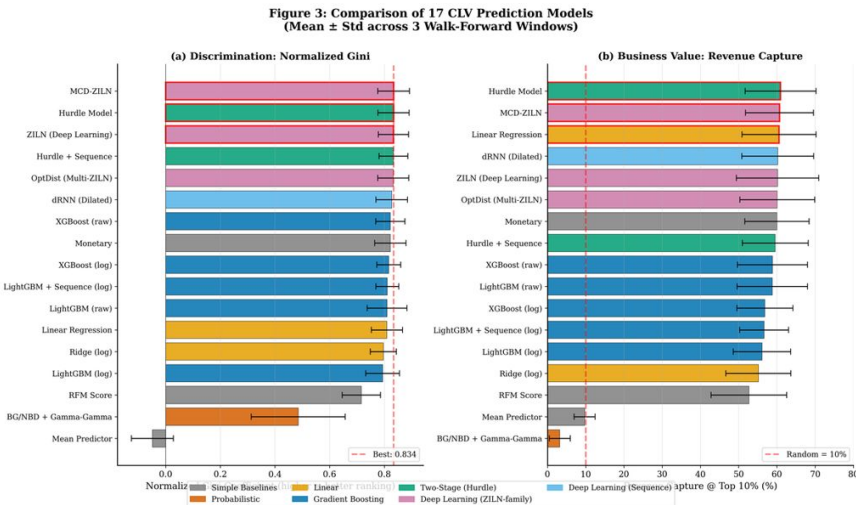
Deep Architectures: Deep ZILN, Optimally Distributed deep nets, dRNN sequences.



Proposed Hurdle Structure

Two-Stage Hurdle: Custom classifier (incidence) combined with specialized regressor (size).

Sequence Extension: Hurdle + deep sequence feature extraction structures.



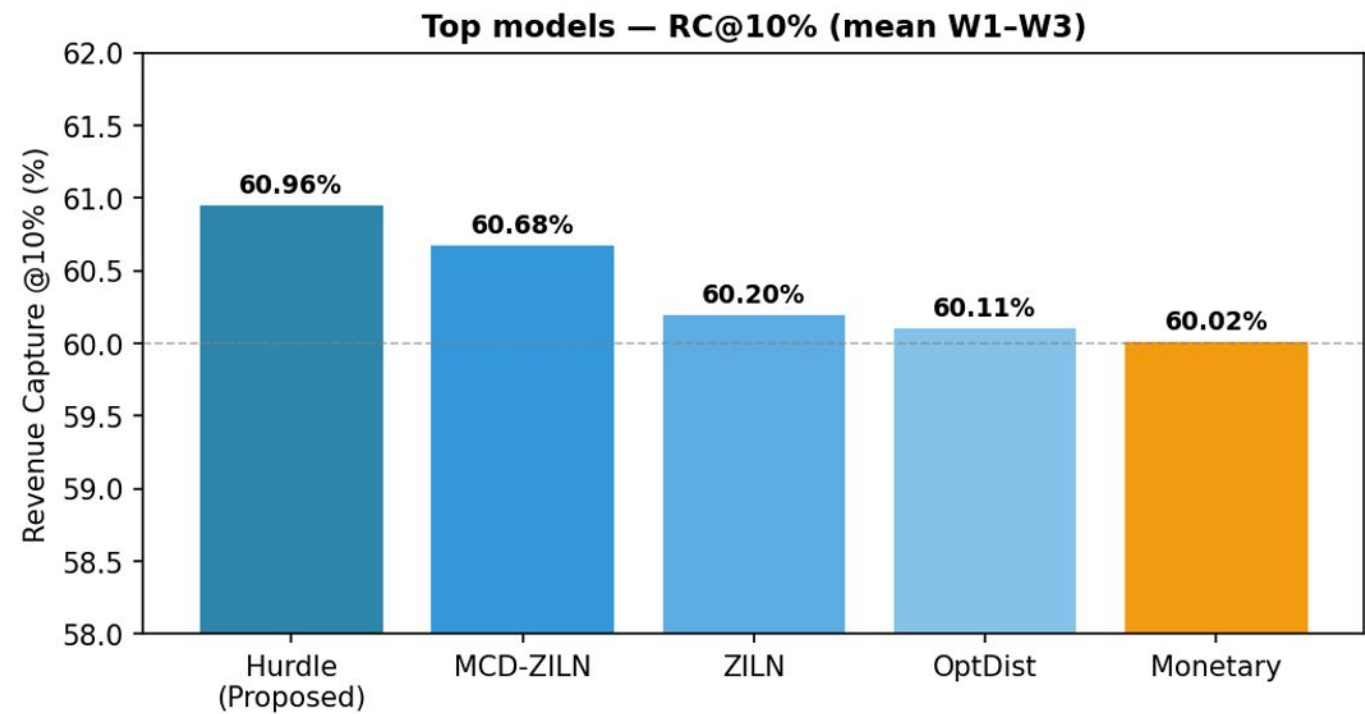
Model Comparison: *Top-K Revenue Capture*

Empirical Interpretation

Rather than claim dramatic performance supremacy, the proposed Hurdle model tracks comparable ranking performance to complex deep-learning alternatives while keeping model weights completely explainable.

NORMALIZED GINI (NG)

Hurdle Model: 0.834 vs MCD-ZILN: 0.834



VIP Marketing Impact & Profit Simulation

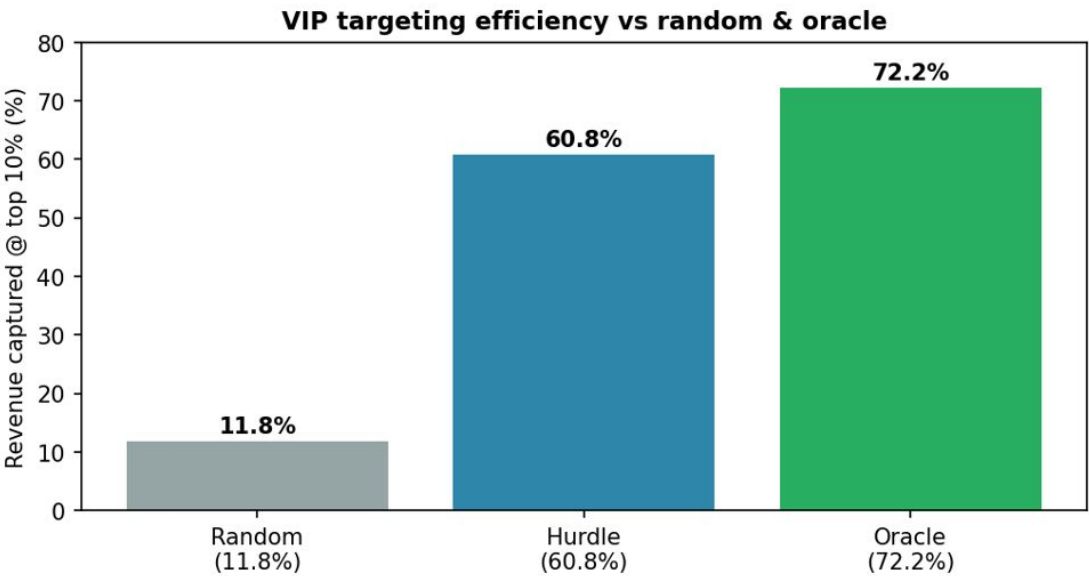


Figure 4: VIP Targeting Performance

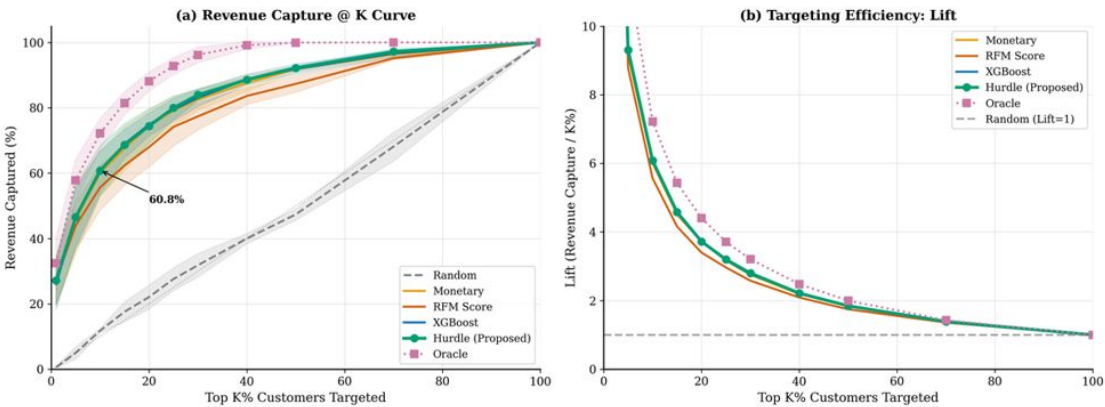
SIMULATED CAMPAIGN ROI

Simulated campaign net profits targeting the top 10% VIP bracket during prediction periods:

Random Selection: \$47,706

Proposed Hurdle: \$292,315

ROI Improvement Lift: +513% vs Random



Separating Decisions: *Stage-Wise SHAP Explainability*

STAGE 1

Will the Customer Return?

Estimates binary active probability **P(Purchase)** within future target periods.

TOP SHAP BEHAVIORAL DRIVERS:

- ✓ **ActiveMonths:** Cumulative engagement density
- ✓ **Recency:** Time elapsed since the last transaction
- ✓ **Sequence Activity:** Normalized short-term transaction rates

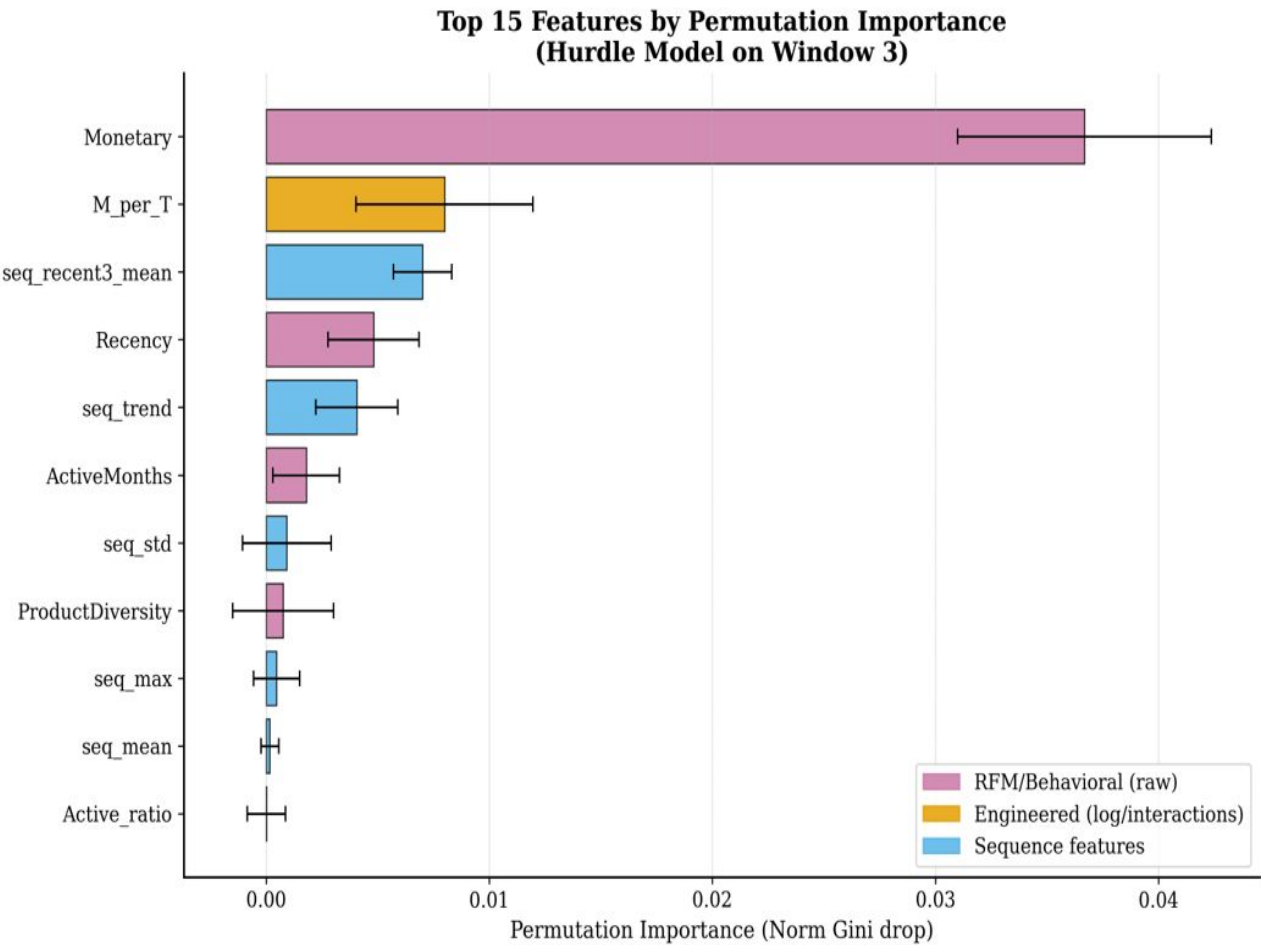
STAGE 2

How Much Will They Spend?

Estimates conditional continuous value **E(Spend | Purchase)**.

TOP SHAP FINANCIAL DRIVERS:

- ➔ **Monetary (M):** Absolute raw financial scale
- ➔ **M/T Ratio:** Average purchase ticket density
- ➔ **Historical Baseline:** Cumulative product affinities

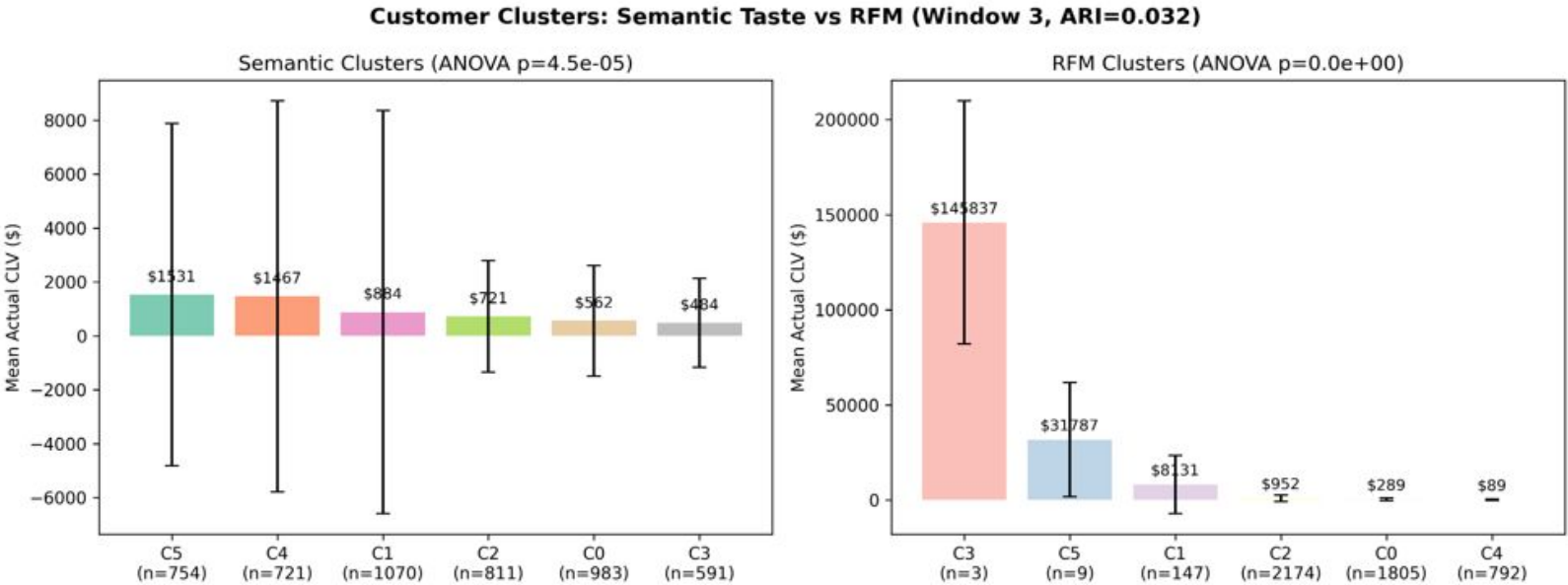


Semantic Features: *Interpretability Over Lift*

- ✓ Product co-purchase graph → semantic taste features
- ✓ Paired tests: $p \geq 0.35$ for NG / RC@10% vs Hurdle-RFM
- ✓ SHAP: semantic = 24.5% Stage-2 importance
- ✓ Clusters orthogonal to RFM (ARI = 0.032), CLV differs ($p = 4.5 \times 10^{-5}$)
- ✓ Claim honestly: segmentation insight, not performance booster.

Stage-2 Feature Significance Rank:

Semantic components make up 24.5% of absolute SHAP feature importance



Reliability: *Calibrated Uncertainty* Bounds

Confidence Strategy	Target Coverage	Average Target Width
Raw MCD-ZILN	34.2%	—
Conformal MCD	94.7%	\$5,729
CQR-Hurdle (Ours)	95.2%	\$1,607

Conclusion: CQR achieves robust target boundary calibration at more than 3.5x narrower intervals.

Conference-Ready Claims

The resulting production system preserves competitive, deep-comparable ranking metrics (0.834 NG) while resolving deployment vulnerabilities.

CORE CORE CONTRIBUTION:

Top-K ranking consistency + local SHAP transparency + semantic customer profiling + calibrated production boundaries.